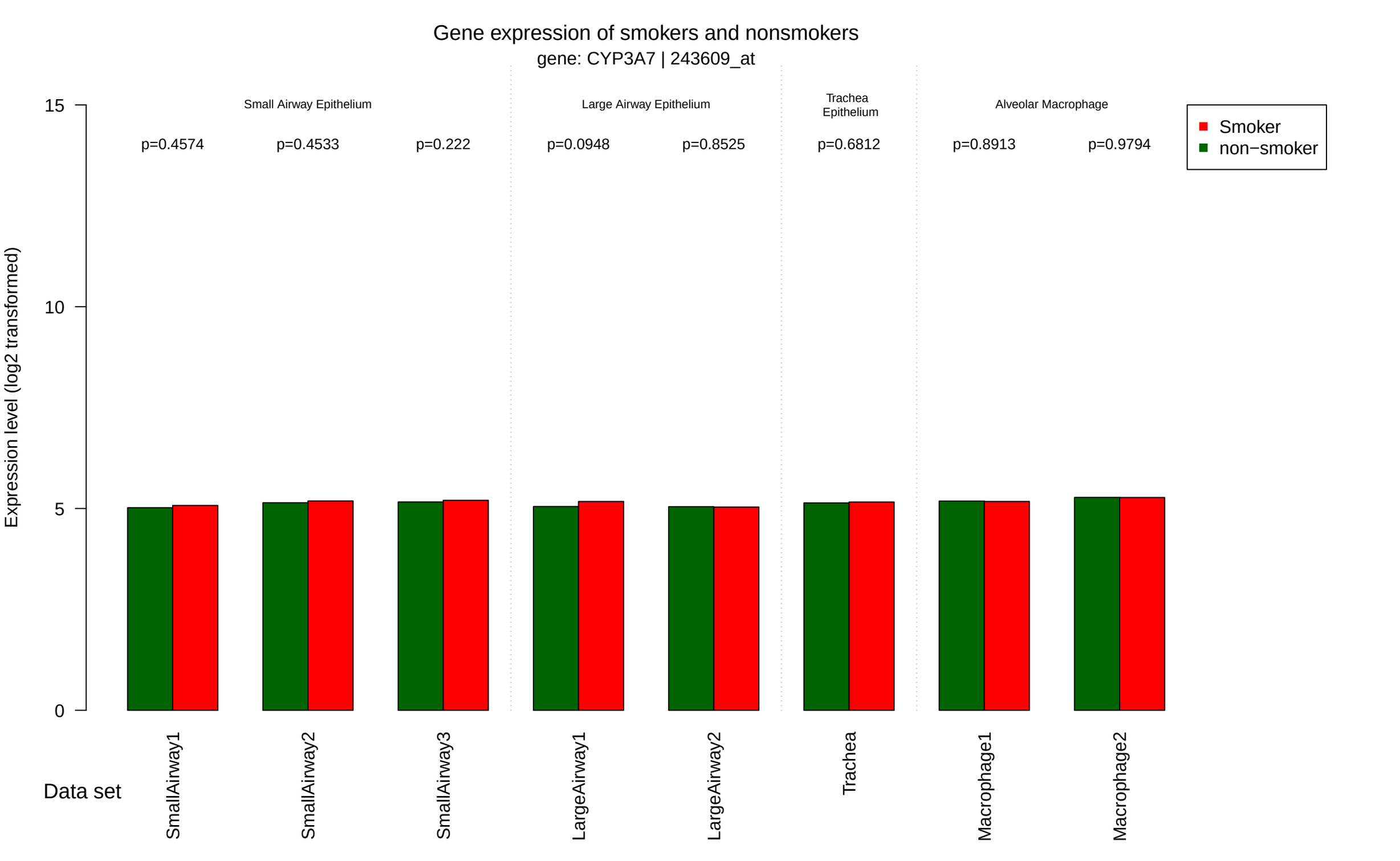
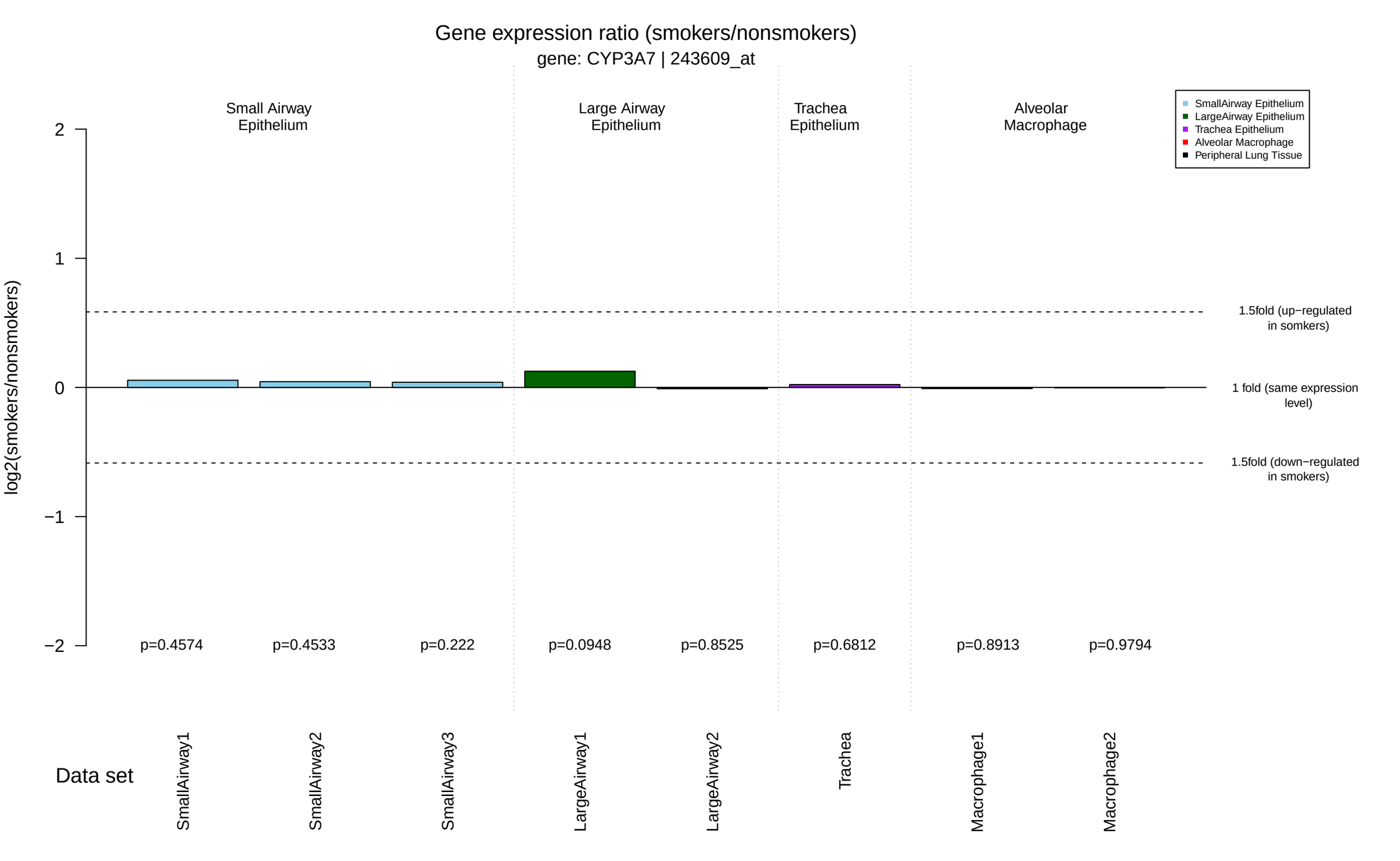
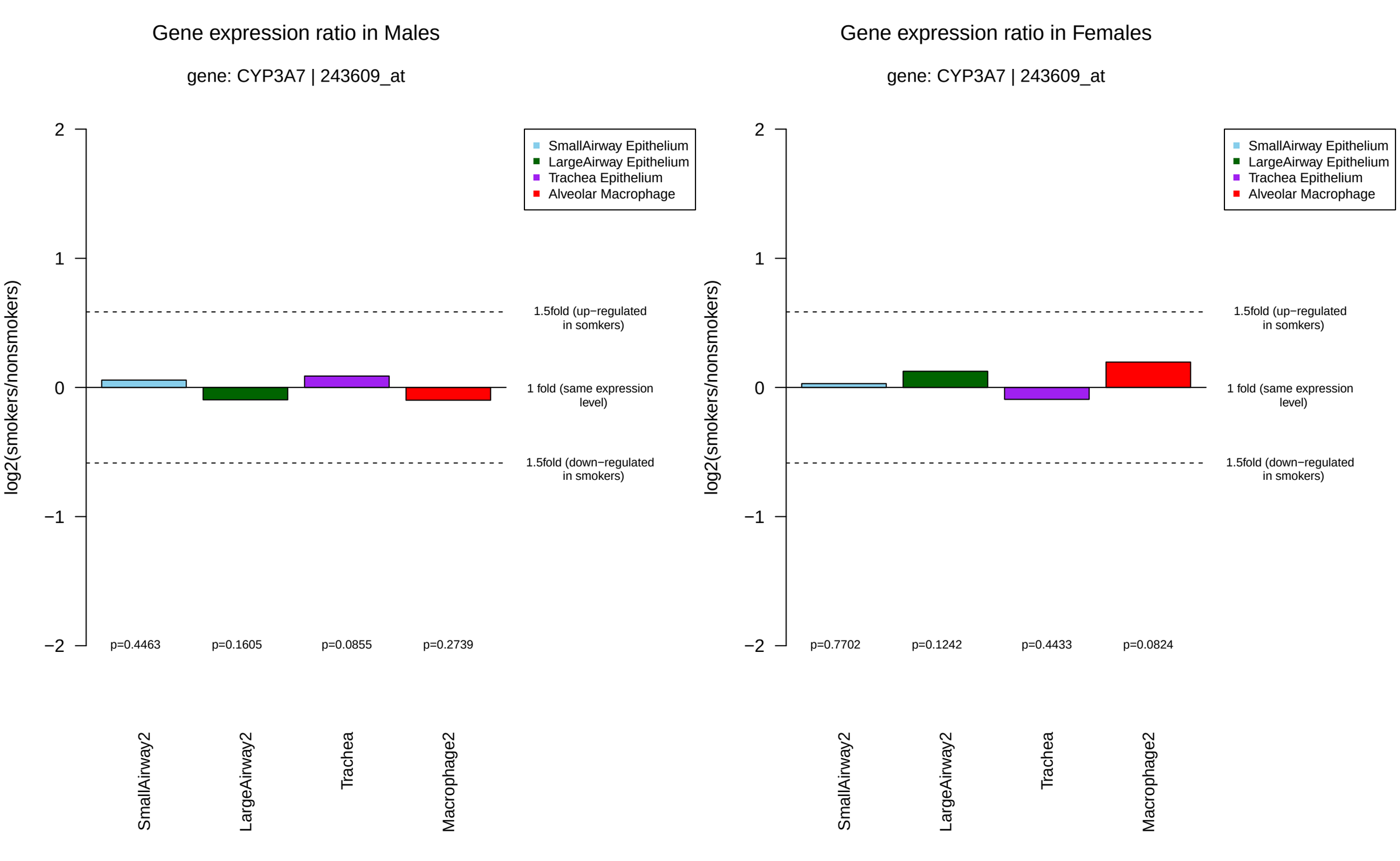
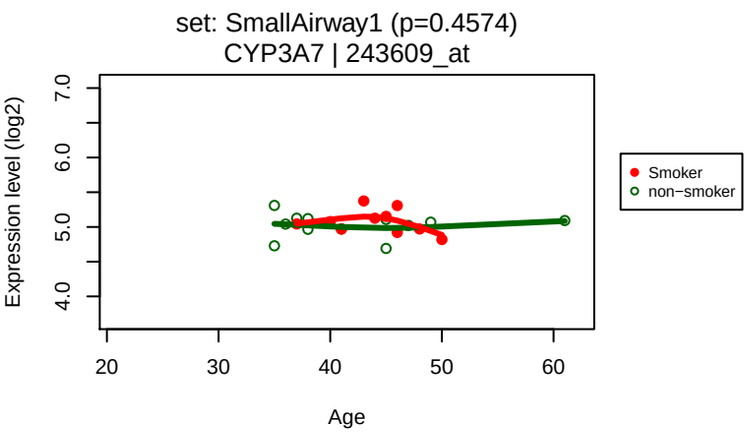
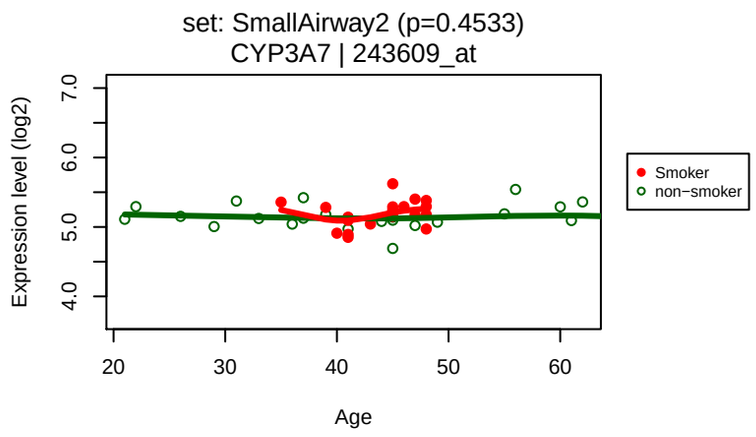
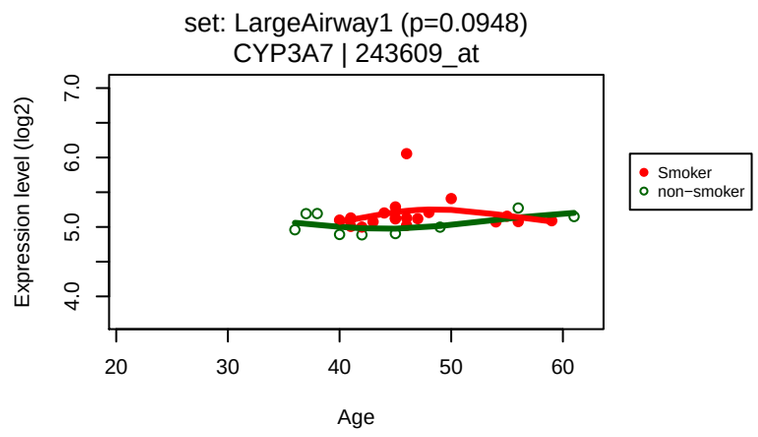

Gene expression of smokers and nonsmokers
gene: CYP3A7 | 243609_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4574
p=0.4533
p=0.222
p=0.0948
p=0.8525
p=0.6812
p=0.8913
p=0.9794
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

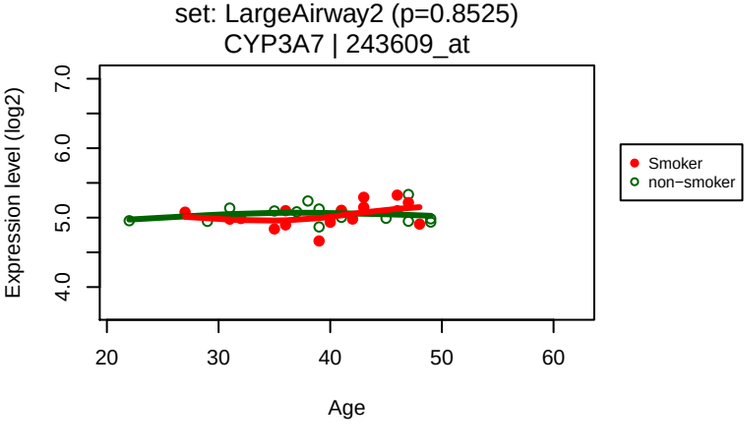

Gene expression ratio (smokers/nonsmokers)
gene: CYP3A7 | 243609_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4574
p=0.4533
p=0.222
p=0.0948
p=0.8525
p=0.6812
p=0.8913
p=0.9794
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

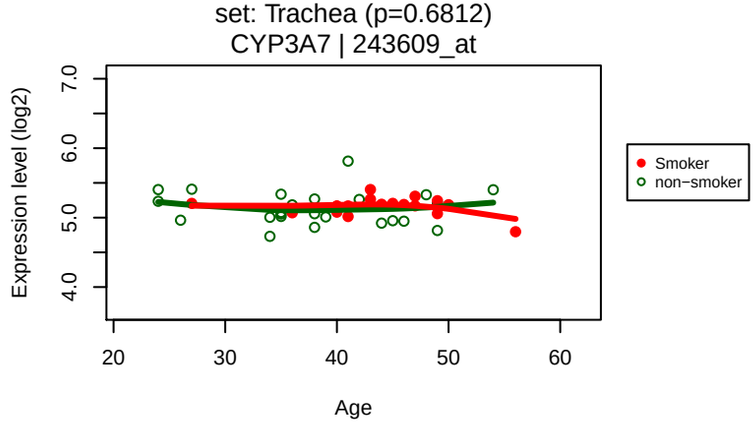

Gene expression ratio in Males
gene: CYP3A7 | 243609_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4463
p=0.1605
p=0.0855
p=0.2739
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP3A7 | 243609_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7702
p=0.1242
p=0.4433
p=0.0824
SmallAirway2
LargeAirway2
Trachea
Macrophage2

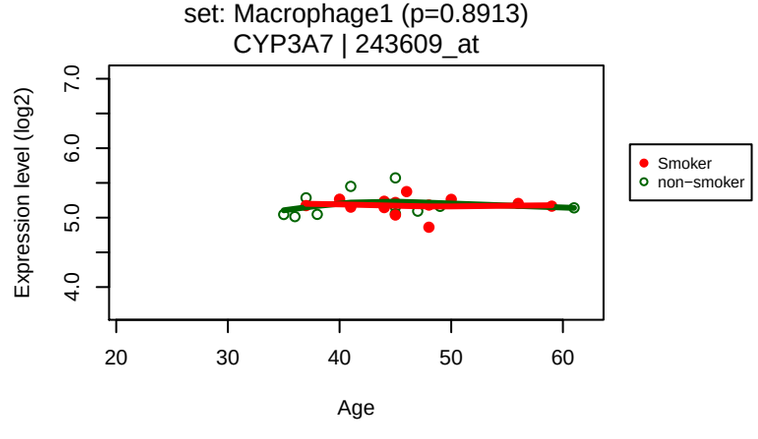

set: SmallAirway1 (p=0.4574)
CYP3A7 | 243609_at
Expression level (log2)
Age
Smoker
non−smoker

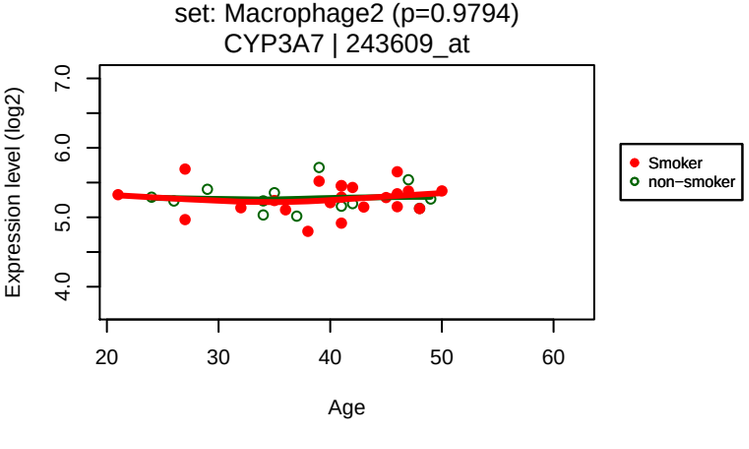

set: SmallAirway2 (p=0.4533)
CYP3A7 | 243609_at
Expression level (log2)
Age
Smoker
non−smoker

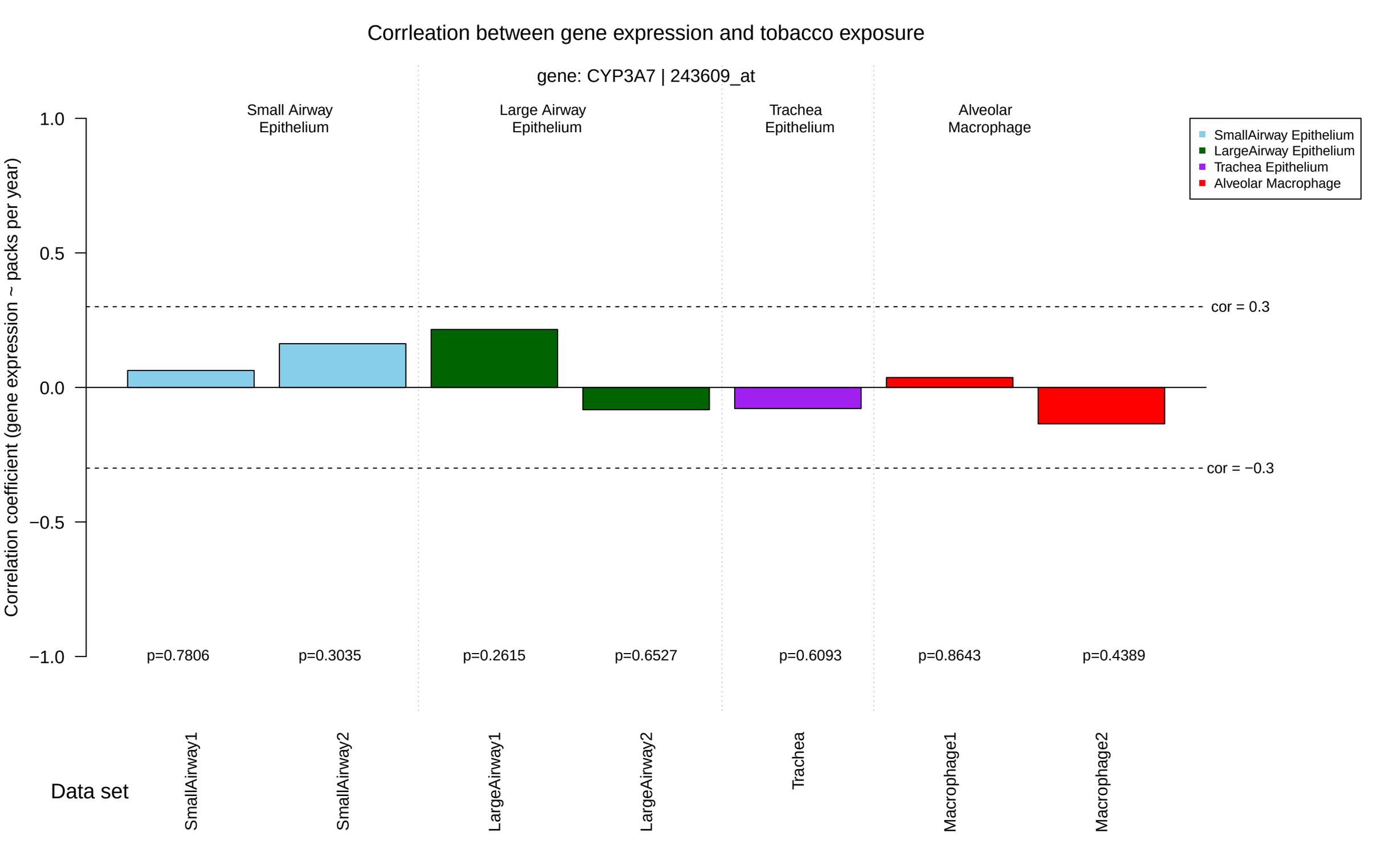

set: LargeAirway1 (p=0.0948)
CYP3A7 | 243609_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8525)
CYP3A7 | 243609_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6812)
CYP3A7 | 243609_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8913)
CYP3A7 | 243609_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9794)
CYP3A7 | 243609_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP3A7 | 243609_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7806
p=0.3035
p=0.2615
p=0.6527
p=0.6093
p=0.8643
p=0.4389
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2